

- 4 A student investigated the structure of muscular arteries.

Figure 4.1 is a photomicrograph of a transverse section (TS) of a muscular artery and surrounding tissue.

Figure 4.2 is a photomicrograph of part of a TS of the muscular artery as indicated in Figure 4.1.

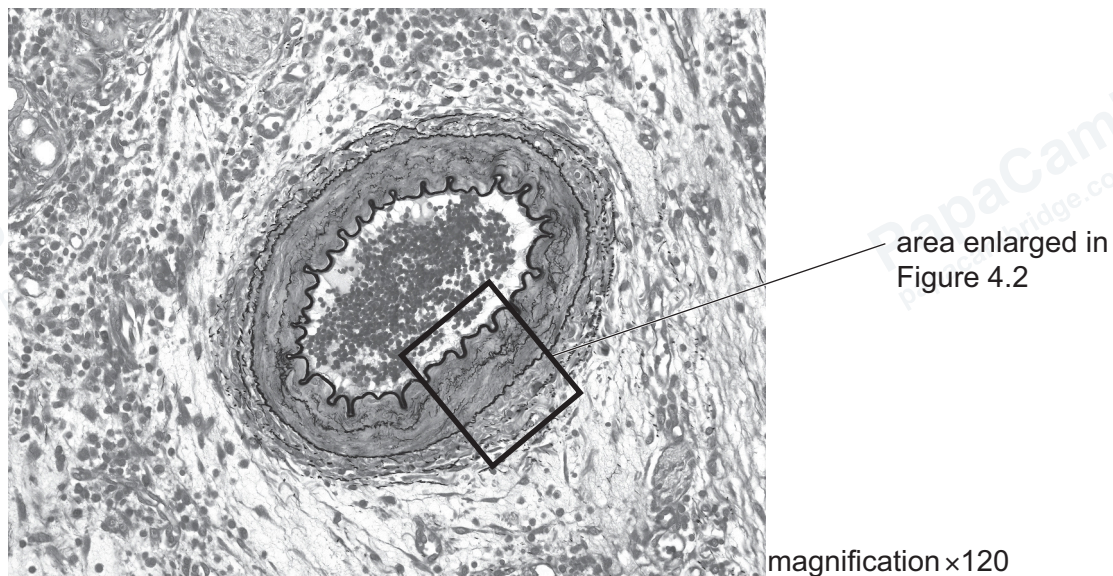


Figure 4.1

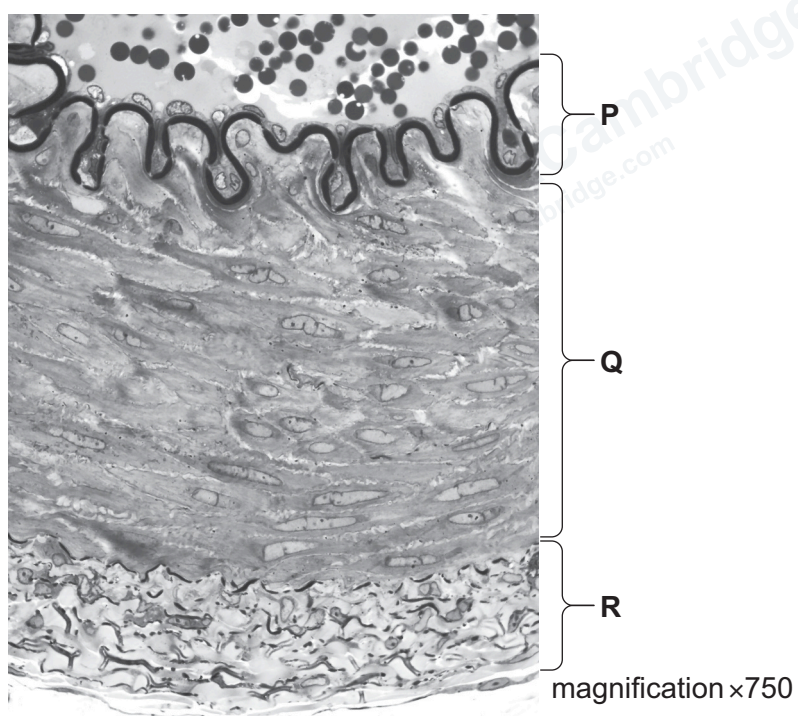


Figure 4.2



(a) (i) Identify the main component of each of the regions **P**, **Q** and **R** shown in Figure 4.2.

P

Q

R

[3]

(ii) Explain how the structure of the muscular artery shown in Figure 4.1 and Figure 4.2 is related to its function.

.....

.....

.....

.....

.....

.....

..... [3]

(iii) Describe how the appearance of a TS through the aorta of a mammal differs from the appearance of the TS of the muscular artery shown in Figure 4.1.

.....

.....

.....

.....

..... [2]



- (b) The student did not observe any neutrophils or lymphocytes in the blood in the centre of the section shown in Figure 4.1.

Draw a neutrophil and a lymphocyte, as seen in a sample of mammalian blood that has been stained to show nuclei.

Do **not** add any labels.

neutrophil

lymphocyte

[2]

- (c) The student also investigated transverse sections of lung tissue.

State **one** feature of the wall of a bronchus that is **not** found in the wall of an artery in lung tissue.

..... [1]

[Total: 11]



Turn over